

Methods: SUNRISE_ARC

ALMA Thesis Planner (automated pipeline)

0.1 0. Week 0: the archive gate, and the three numbers that fix everything downstream

Before you image anything, before you even download anything large, spend a few hours in the ALMA Science Archive on the project code named in the acknowledgement, 2023.1.00802.S. Pull up the project page and the delivered data products, and record, in a lab notebook file that you will cite in the thesis: (i) the observation status of each scheduling block and whether both the compact and extended configurations were executed and passed QA2; (ii) the proprietary-expiry date; (iii) the delivered synthesized beam (major and minor axis, position angle) for each configuration and each product; (iv) the spectral setup — every spectral window, its centre frequency, bandwidth, and native channel width; and (v) the achieved RMS per channel per beam from the QA2 report and the pipeline weblog, together with the channel width at which that RMS is quoted.

Every number that later sizing depends on comes from this page and the QA2 products — never from the as-proposed configuration in the proposal. The proposal tells you what was asked for; the weblog tells you what the array delivered. Where they disagree, the weblog wins, and you say so in the thesis.

From those, you compute and freeze three quantities before any science analysis:

- **σ_chan** — the achieved RMS per channel per beam, converted to a common channel width (I suggest you carry it at both the native width and at a $30\text{--}50\text{ km s}^{-1}$ binned width, since the binning choice is yours later and you want the scaling written down now: σ scales as the inverse square root of the binning factor for independent channels).
- **N_beam** — the number of independent beams along the $\sim 16''$ arc. Compute it as $16''$ divided by the beam FWHM projected along the arc's local direction, not along the beam major axis by default; if the beam is elongated, work out the effective resolution along the ridge for the actual arc position angle and record both.
- **The per-beam velocity-centroid threshold.** The centroid uncertainty of a line of width FWHM measured at signal-to-noise SNR is approximately $FWHM/(2.35 \cdot SNR)$. Invert this: for a plausible dwarf-galaxy [CII] line width of a few tens of km s^{-1} , ask what SNR you need for a centroid precision that is a meaningful fraction (say a third) of that width. Then, using σ_chan and the delivered beam, compute what integrated line flux per beam is required to clear that SNR in a moment-0 over a line window of that width, and write down how many independent positions along the arc could plausibly clear it. Adopt the hard rule now: **a velocity centroid is quoted only where the line is detected at $\geq 5\sigma$ in the moment-0 integrated within that aperture.** Positions below that are reported as non-detections with limits, never as noisy velocities.

0.2 1. Pre-specify the SNR ladder, and declare the rung before you look at the data

The core kinematic measurement in §6 will be executed at whichever rung the week-0 arithmetic supports. Write the rung choice down, dated, before the first cube is made:

- **Rung A** — per-beam spectra along the ridge line; a full position–velocity (PV) diagram; per-beam velocity-centroid and line-width profiles.
- **Rung B** — centroids in N coarse bins along the arc, with N set so each bin reaches the threshold SNR (typically 3–5 bins rather than ~ 15 beams). A three-bin blue/middle/red measurement across a $16''$ arc is a real kinematic result, and it is the expected outcome if the integrated line lands near the proposal’s design sensitivity.
- **Rung C** — the integrated line only, plus a red-half/blue-half velocity difference and a stacked-spectrum test for asymmetry or a second component. One number, but a measurement rather than a null.

There is no rung at which §6 produces nothing. The rung is provisionally set in week 0 from the predicted sensitivity and is **locked** in §5, after the integrated detection significance is actually measured.

Contingency, decided at the gate, not in month six. If the archive shows the data are still proprietary beyond your thesis window, the project converts at week 0 to a methods-and-forecast thesis: you build and validate the complete extraction and injection–recovery pipeline of §§4–9 on CASA-simulated cubes matched to the as-proposed configurations, produce an explicit sensitivity forecast for what the delivered data will measure, and write a synthesis of published work on this arc. That is a reduced scope, stated as reduced, and it is a decision made in week 0 with your advisor — not a rescue discovered later.

0.3 2. Frequency and measurement provenance, settled from in-hand data only

The data description labels the line “[CII]158um” but does not state whether that 158 μm is rest-frame or observed-frame, and quotes no frequency at all. Carry it in the thesis as an **unverified wavelength label**, and say that explicitly in your methods chapter.

Do the arithmetic both ways as a sanity check you can perform with no external input. If 158 μm is a *rest-frame* label, the expected observed-frame frequency is $\nu_{\text{obs}} = c/(158 \mu\text{m} \cdot (1+z))$ with $z = 5.93$ as NIRSpec-confirmed, i.e. roughly 270–275 GHz — the top of ALMA Band 6 or the bottom of Band 7. If instead the 158 μm were an *observed-frame* wavelength, the implied frequency is near 1.9 THz, which lies outside every ALMA receiver band; since the observations exist, that reading is excluded. This is a self-consistency argument from the dataset, not a verification of the label, and you should present it that way.

The authoritative record is then the correlator setup itself. Run ‘listobs’ on the retrieved measurement sets, and inspect the spectral windows with ‘plotms’ and with the ‘analysisUtils’ spectral-setup helpers. Where the observers centred a spectral window relative to your derived ν_{obs} tells you where they expected the line, and the remaining windows tell you which channels are line-free for continuum. **No external line database is consulted** — none is named in the description, and none is needed, because the correlator setup is the self-contained record for this dataset.

Carry the redshift as “ $z = 5.93$, NIRSpec-confirmed,” not as an exact adopted constant. Keep

the line-search window wide — at least $\pm 1000 \text{ km s}^{-1}$ about the expected ν_{obs} , limited by the spectral window edges — so that a genuine velocity offset between [CII] and the rest-UV/optical features that set the NIRSpec redshift cannot be flagged out by construction. Set ‘restfreq’ in ‘tclean’ to your derived rest frequency, adopt ‘outframe=LSRK’, and state both choices; every velocity in the thesis is then explicitly “relative to $z = 5.93$ in the LSRK frame.”

Two more provenance rules, fixed now. The magnification is quoted in the description as an approximate $\mu \sim 300$ **for Earendel specifically**, a highly localized value near the critical curve; it is never applied to the arc as a whole, and it never appears in any arc-integrated number. The clump radii “between 1 and 20 pc” are **NIRCam-inferred source-plane sizes**, carried as such throughout, and never presented as ALMA-measured sizes.

0.4 3. Retrieval, calibration, and training — weeks 1–10

Weeks 1–3 are explicit CASA training and are not science. Work through the delivered pipeline weblog page by page: the calibrator amplitude and phase solutions, the bandpass, the flagging summary, the per-EB sensitivity tables, the QA2 imaging products. Learn to read a weblog before you learn to run ‘tclean’. In parallel, run the standard CASA guides on a delivered calibrated dataset so that ‘listobs’, ‘plotms’, ‘flagdata’, ‘tclean’, ‘imstat’, ‘imfit’, and ‘immoments’ are muscle memory before you touch the science target.

Then retrieve the data. **If the archive offers delivered calibrated products (calibrated measurement sets, or the ‘scriptForPI’-style restore from the pipeline), take that path and say so in the thesis.** The value of this thesis is not in re-deriving a calibration that the observatory pipeline already produced and QA2-verified. Use ‘scriptForPI.py’ or the pipeline restore task to regenerate the calibrated MS from the raw ASDMs plus the delivered calibration tables, in the CASA version named in the weblog — version mismatches are the single most common source of a failed restore, so pin the version.

Once restored: run ‘listobs’ and record the number of execution blocks, on-source time, and antenna configuration for each. Inspect the applied flags with ‘flagdata(mode=‘summary’)’ and note the flagged fraction per spectral window and per antenna. Re-inspect the bandpass and phase-calibrator solutions with ‘plotms’; you are not looking to improve them, you are looking to be able to state in the thesis that you checked them and they are sound. Check the flux-calibrator model and the derived flux scale, and quote the ALMA systematic flux-calibration uncertainty for the band in question as a separate term in your final error budget, added in quadrature to the statistical error on every flux you report.

The compact configuration is the primary dataset throughout. Concatenate the compact-configuration execution blocks with ‘concat’ after confirming their spectral setups match, and keep the extended configuration in a separate MS, untouched until §10.

0.5 4. Continuum and cube imaging

Line-free channel selection. Using the derived ν_{obs} and a generous $\pm 1000 \text{ km s}^{-1}$ exclusion zone, define the line-free channel ranges across all science spectral windows. Record the selection explicitly as a CASA ‘spw’ string in the thesis appendix.

Continuum image. Run ‘tclean’ with ‘specmode=‘mfs’’, ‘deconvolver=‘mtmfs’ with ‘nterms=1’ (a single-band continuum cannot constrain a spectral index, so do not fit one), ‘weighting=‘briggs’ with ‘robust=0.5’ as the fiducial, ‘gridder=‘standard’ unless the delivered data are mosaicked, and

clean shallowly to $\sim 3\sigma$ inside a hand-drawn mask following the arc, plus a deep unmasked run for comparison. Produce a primary-beam-corrected image and keep the uncorrected image for noise measurement (noise is flat only in the uncorrected map). Repeat at ‘robust=2.0’ (natural) and with one uv-taper chosen to roughly double the beam area, since surface-brightness sensitivity, not point-source sensitivity, is what matters for a $16''$ arc.

Continuum subtraction and the cube. Fit and subtract the continuum in the visibility domain with ‘uvcontsub’, ‘fitorder=1’, using the line-free channel selection above; verify by re-imaging the line-free channels of the subtracted MS and confirming they are consistent with noise. Then image the [CII] cube with ‘specmode=cube’, ‘outframe=LSRK’, your derived ‘restfreq’, ‘deconvolver=multiscale’ with scales of roughly [0, 1, 3, 9] times the beam in pixels (the arc is extended; a point-source-only deconvolution will underclean it), ‘weighting=briggs’ ‘robust=0.5’, and ‘usemask=auto-multithresh’ with the standard ALMA compact-configuration parameters as a starting point, cross-checked against a hand mask. Make the cube at three channel binnings — native, $\sim 30 \text{ km s}^{-1}$, and $\sim 50 \text{ km s}^{-1}$ — plus a tapered version at the intermediate binning. Pixel size should give 4–5 pixels across the beam minor axis.

Measure the achieved RMS per channel in the non-primary-beam-corrected cube with ‘imstat’ over signal-free channels and signal-free sky regions, and compare it to the week-0 σ_{chan} and to the theoretical sensitivity. Any discrepancy larger than $\sim 20\%$ is a flag to go back to the weblog before proceeding.

0.6 5. Result #1: integrated [CII] and continuum over the full arc — weeks 10–15

This measurement is self-contained and stands regardless of everything that follows.

Aperture definition. Define the arc aperture from the data themselves. Build a preliminary moment-0 by integrating the cube over a $\pm(1.5 \times \text{expected FWHM})$ window about the expected ν_{obs} , smooth it to twice the beam, and draw a polygonal region enclosing the emission; where no emission is evident, define the aperture from the continuum image, and if neither shows structure, define it geometrically from the $\sim 16''$ arc morphology described in Welch+22. Do a **curve of growth**: extract the integrated spectrum in a sequence of nested apertures of increasing area and plot integrated flux versus aperture area. Adopt the aperture at which the curve flattens; if it never flattens, adopt the largest aperture whose integrated SNR is still rising and say so. Record the adopted aperture area in beams — you need it for the noise scaling.

Integrated spectrum and line fit. Extract the aperture-integrated spectrum with the primary-beam-corrected cube, converting from Jy beam^{-1} to Jy using the aperture-to-beam area ratio. Fit the line with (a) a single Gaussian and (b) a double Gaussian, using ‘emcee’ with flat priors on amplitude, centroid, and width, and compare the two with the Bayesian Information Criterion; report the single-Gaussian result as fiducial unless the two-component model is preferred by $\Delta\text{BIC} > 10$. Deliverables: integrated flux $S_{\text{[CII]}}\Delta v$ in Jy km s^{-1} (or a 3σ upper limit), FWHM in km s^{-1} , and velocity centroid in km s^{-1} relative to $z = 5.93$.

Non-detection branch. If the integrated line is not detected at $\geq 5\sigma$, compute a 3σ upper limit as $3 \cdot \sigma_{\text{chan}} \cdot \sqrt{N_{\text{chan}}} \cdot \Delta v_{\text{chan}} \cdot \sqrt{A_{\text{aperture}}/A_{\text{beam}}}$, for an assumed line width you state (quote limits for 50, 100, and 200 km s^{-1} so the reader can rescale). Also run a blind line search over the full usable bandwidth within the aperture — a matched-filter search over a grid of trial widths and centres — and report the significance of the highest peak alongside the

distribution of negative peaks, which gives you an empirical false-positive rate.

Continuum. Measure the continuum flux in the same aperture with ‘imstat’ and, if the source is compact enough, with ‘imfit’; report a flux or a 3σ limit set by the aperture-scaled RMS. Add the ALMA flux-scale systematic in quadrature.

Derived apparent quantities. Compute the apparent [CII] luminosity via the standard relation $L = 1.04 \times 10^{-3} \cdot S_{\text{[CII]}} \Delta \nu \cdot \nu_{\text{obs}} \cdot D_L^2$, adopting and stating a flat Λ CDM cosmology ($H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$) and D_L at $z = 5.93$. Report it as $\mu \cdot L_{\text{[CII]}}$, an apparent, magnification-scaled quantity; intrinsic values are deferred and flagged as requiring a lens model outside this dataset. For the continuum, a single band cannot constrain the dust SED, so compute apparent L_{IR} over an explicitly stated **grid** of modified-blackbody assumptions: dust temperature $T_d = 35, 45, 55, 65 \text{ K}$ crossed with emissivity index $\beta = 1.5, 1.8, 2.0$, including the CMB-at- $z=5.93$ correction to both the dust heating and the observed contrast against the background. Present the result as a table over that grid, never as a single number. Any obscured SFR you quote is $\mu \cdot \text{SFR}$ and carries the same grid.

Lock the rung. The measured integrated SNR here confirms or demotes the week-0 rung choice. Recompute N_{beam} and the per-beam threshold using the *measured* line width and the *measured* per-beam RMS rather than the assumed ones, and lock the rung. Write the locked rung and the arithmetic behind it into the thesis before starting §6.

0.7 6. The thesis core: the velocity cut along the arc — weeks 15–26

Trace the ridge line. From the moment-0 (or, if the line is weak, from the continuum, or failing both from the arc geometry described in Welch+22), extract the emission ridge: for a sequence of cuts perpendicular to the local arc direction, find the flux-weighted centroid of the transverse profile, then fit a smoothing spline through those centroids. Parameterize the ridge by arc length s in arcseconds from one end. The tracing is done on a beam-smoothed map to avoid chasing noise, and you should show the traced ridge overlaid on the map as a figure. **You require no clump coordinate list at any point** — the description names no paper or catalog tabulating clump positions, so the entire analysis is built to work without one.

Rung A execution. Extract spectra in circular apertures of diameter one beam FWHM, centred on the ridge, spaced by one beam (independent samples). Fit each with a single Gaussian; retain the centroid and width only where moment-0 SNR ≥ 5 in that aperture. Build the PV diagram with CASA ‘imprv’ along the traced ridge (‘imprv’ takes a straight slit, so for a curved ridge either piece it together from straight segments or extract the PV array manually by interpolating the cube onto the ridge path), using a slit width equal to the beam FWHM. Present the PV with contours at 2, 3, 4, 5σ , where σ is measured empirically from the same extraction on blank sky.

Rung B execution. Partition the ridge into N contiguous arc-length bins, N set so each bin’s integrated moment-0 clears the threshold SNR. Extract and fit one spectrum per bin. Report centroid and FWHM per bin with uncertainties, and show the binned PV.

Rung C execution. Split the arc at its midpoint in arc length into a “red half” and “blue half” *defined geometrically, before looking at the velocities* — this matters, because a split chosen after seeing the data manufactures a gradient. Extract and fit both spectra, and report $\Delta v = v_{\text{half1}} - v_{\text{half2}}$ with its uncertainty. Additionally, stack the full-arc spectrum and test for asymmetry (fit a Gauss–Hermite h_3 term, and separately compare a one- versus two-component fit by BIC) as the test for a second velocity component.

Uncertainties come from the sky, not from the fitter. This is the rule I care most about. For every quantity you measure — integrated flux, per-position centroid, per-position width, Δv — repeat the *identical* extraction, with the *identical* aperture shape and size and the *identical* fitting procedure, at ≥ 200 positions on emission-free sky within the primary beam and away from the arc, and at the same set of positions in line-free channel ranges. The scatter of those recoveries is your uncertainty. Quote formal fit errors separately and note where the two disagree; where they disagree, the empirical value is the one you use. Where a fit fails to converge on blank sky, that failure rate is itself a reported number.

Null tests, stated in advance. Test three propositions, each against an explicit null: 1. *Is there a monotonic velocity gradient along the arc?* Fit $v(s) = v_0 + (dv/ds) \cdot s$ by weighted least squares; compare to the flat-velocity null ($dv/ds = 0$) by $\Delta\chi^2$, and calibrate the significance by permutation — randomly reshuffle the measured centroids among the positions $\geq 10^4$ times and record how often the shuffled data produce a $|dv/ds|$ as large as observed. Report dv/ds with a confidence interval and the permutation p-value. Also report Spearman’s rank correlation between v and s , which does not assume linearity. 2. *Are there distinct velocity components?* Per position and in the stack, compare one- versus two-Gaussian fits by BIC, with the ΔBIC threshold calibrated on the blank-sky extractions so you know what ΔBIC noise alone produces. 3. *Does the line width vary systematically along the arc?* Fit $\text{FWHM}(s)$ linearly and against a constant, with the same permutation calibration.

Report all three with confidence intervals whatever the outcome. A confidently measured *absence* of a gradient, with a quantitative bound on dv/ds , is a result and is written up as one.

0.8 7. Interpretation, with the lensing caveat carried explicitly — weeks 26–29

No lens model is named in the data description, so **every result in this thesis is stated as an image-plane observable**. Write your results chapter with that discipline built into the sentence structure: “the image-plane [CII] surface brightness along the arc,” “the image-plane velocity centroid at arc-length s .”

Two inferences are explicitly withheld, and you should state the withholding rather than leaving it implicit. First, smooth image-plane emission does not imply diffuse source-plane gas: differential magnification along a fold can smear genuinely clumpy emission into an apparently continuous ridge. Second, an image-plane velocity gradient does not imply source-plane rotation: mirrored images across the critical curve can mimic an ordered gradient where none exists, or fold an existing one into apparent symmetry. Both inferences are deferred pending a lens model, which is outside this dataset.

What the image-plane result *does* support with no model at all is quantitative follow-up design, and that is how you frame the payoff. The observed surface-brightness contrast and velocity contrast on the sky are exactly the quantities a long-baseline program must detect; your achieved per-beam, per-channel sensitivity converts directly into required integration times at any specified higher angular resolution, via the standard scaling of point-source and surface-brightness sensitivity with beam area and integration time. That framing is magnification-agnostic, so state it plainly and carry it into §8.

0.9 8. Injection–recovery on the real visibilities — weeks 29–34

Run exactly **one** injection campaign, and run it in the visibility domain on the real data at the true delivered beam and true delivered RMS. No separate simulated-cube campaign.

Construct two families of model cubes on the traced ridge line, in image coordinates matched to your science cube:

- a **clump-following template**: a series of unresolved (beam-scale in the image plane) components placed along the traced ridge at a grid of spacings and a grid of clump-to-inter-clump surface-brightness contrasts, each given the velocity structure measured or bounded in §6;
- a **constant-along-arc template**: the same total flux distributed uniformly along the ridge.

For each model, use ‘ft’ to fill MODEL_DATA in a copy of the calibrated compact-configuration MS with the model cube, then add it to the data with ‘uvsub(reverse=True)’, and re-image with the *identical* ‘tclean’ call used for the science cube. Then run the *identical* ridge extraction and fitting of §6 on the injected cube, and measure the recovered flux fraction, the recovered clump contrast, and whether the injected clump spacing is detectable above the empirically calibrated noise.

The output is a recovered-contrast curve: as a function of clump spacing and injected contrast, the fraction of injections in which the clumpy model is distinguishable from the constant model at a stated significance. Express the result as a **design requirement** — the beam size, per-channel sensitivity, and configuration needed to separate clumps from diffuse gas along this arc at $z = 6$. This yields a quantitative number regardless of whether the science line is detected, and it is a chapter, not the thesis.

As a short exploratory add-on only, search the 1D brightness and velocity profiles $v(s)$ and $I(s)$ for reflection symmetry about a free-position axis, by fitting the axis position that minimizes the residual between the profile and its own reflection, with the significance calibrated by the same permutation machinery. **Report this strictly as a bound, with the caveat stated in the text:** fitting a free axis position in a low-SNR 1D profile is weakly constrained, and the axis being fitted is effectively the critical-curve position, so this is not a model-free test and must not be presented as one.

0.10 9. Bounded use of the extended configuration

The extended-configuration data are **not** imaged clump-by-clump. Inspect them only far enough to answer one question, and only if §§3–6 finish on schedule: is a significant fraction of the [CII] flux resolved out on long baselines? Do this by imaging both configurations restricted to a common uv range, comparing the integrated arc flux in that matched range, and separately by plotting the real part of the visibility amplitude versus uv distance, azimuthally averaged and phase-referenced to the arc centre, for the combined dataset. Report the missing-flux fraction with its uncertainty, or state that the comparison was not reached. Clump-by-clump imaging is recorded in the thesis as a deliberate scope choice made at the outset, not as something the data failed to support.

0.11 10. Reproducibility and the written record

Keep every ‘tclean’ call, region file, channel selection, and fitting script under version control, with the CASA version pinned, and include the full parameter sets in an appendix. Every figure should be regenerable from a single script given the calibrated MS. Carry the flux-calibration systematic, the

empirical statistical error, and the dust-SED grid as three separately labelled sources of uncertainty in every table — never merged into one error bar. And in the methods chapter, state in your own words the four provenance rules you have been following: the 158 μm label is unverified as to frame; $z = 5.93$ is NIRSpec-confirmed and carried as such, not as an exact constant; $\mu\sim 300$ is an approximate, Earendel-local value and is never applied to the arc; and the 1–20 pc clump radii are NIRCам-inferred source-plane sizes, not ALMA measurements.

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